Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

1. Tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and use their own software and AI to make them faster and with fewer skilled people than older competitors. They earn money on what they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product. The takeaways below cover why demand is rising, what makes these businesses hard to build, and where the money ends up.

1. **Some tailwinds behind US production are lasting and others can reverse. Build the thesis on the lasting ones.**
   - **Lasting:** China's control of mineral refining and magnets, the shortage of skilled factory workers, and the years it takes a big buyer to approve a new supplier. None of these goes away with an election or a budget cycle.
   - **Temporary:** tariffs, the current surge in munitions spending and interest rates. Tariffs are set by proclamation and can change overnight. Munitions money depends on bills Congress has not passed. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR).
   - **Implications:** favor companies that still work if tariffs fall and the munitions surge slows. That means owning approved capacity that buyers need for a decade, and avoiding businesses whose margin exists only because of a tariff.
2. **China's hold on minerals and magnets is the most lasting limit in the theme.**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA). Refining is the step that turns ore into pure metal. China also makes about 94% of the world's rare-earth magnets (Mining.com). These magnets go into drones, missiles, motors and robots.
   - The US is fully import-reliant for 16 critical minerals (USGS). China has already used export controls on these materials, and its current pause on those controls ends on November 10 and November 27, 2026.
   - This is why buyers and the government will pay for a non-Chinese supply even when it costs more. It also means the plants that fix this problem are expensive and slow to build.
3. **Defense demand and government money are real, but much of the money is conditional and slow.**
   - The FY27 budget request asks for $49.5B for 10 key munitions, about 8.5 times the past decade's average. About 82% of that depends on a reconciliation bill Congress has not passed (CNAS). The Navy is putting $900M into plants that make submarine parts (Breaking Defense).
   - The FY26 defense bill lets the Pentagon buy munitions over several years and commit to purchases of up to 10 years, with advance payments. That is new and helpful, though each year still needs Congress to approve the money (FY26 NDAA).
   - Government now acts as lender and investor. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to one magnet maker and a $500M conditional loan to a rare-earth metals maker, and about $20B is requested for FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE). Governments worldwide put about $65B into minerals in 2025 (IEA). These loans take a long time to close and come with conditions.
4. **The labor gap is lasting, and it is driving owners of small shops to sell.**
   - US manufacturers need to fill about 3.8M jobs by 2033, and about 1.9M are projected to go unfilled (Deloitte).

- Of 17,156 US machine shops, 83% have fewer than 20 employees (Census). Many owners are near retirement with no one to take over. That creates a supply of certified shops for sale at reasonable prices.
- Fewer skilled people means a plant that needs fewer of them can grow where others cannot. That is the opening for software and automation.

5. **Capex before contracts is the central risk. A factory has to spend its money before it wins the orders.**
   - Capex means capital spending on buildings, machines and certifications. A big defense contractor, often called a prime, won't approve a new supplier or sign a volume contract until the plant exists and is certified. Approval takes about nine months to two years: one prime told us 9-12 months in an expert call, and another said one to two years.
   - One precision-parts maker committed about $200M to a single Arizona plant (Plant Services), and its CEO says capex runs about two years ahead of revenue (claim). Process plants need about 80-85% use of capacity to make money (FTI Consulting). A hardware company at Series B often has about $10M of revenue and break-even gross margins (Lowercarbon).
   - This hurts venture returns. Equity spent on a building earns building-like returns and dilutes founders before the first volume order arrives.
   - The winners shift who pays for the build. Options include government loans and grants, a buyer that agrees to purchase output before the plant is built (the Pentagon agreed to buy all of one magnet plant's output for 10 years, with a price floor), a customer that hosts the factory inside its own plant (one 7-year deal), bank debt once revenue exists, leased buildings, franchising, and buying plants that are already certified. One roll-up founder said they buy shops at about seven times annual earnings (claim).
6. **Buyers pay for speed and spare capacity. Few pay for a lower price, so the margin goes to whoever owns the output and holds the approvals.**
   - "Cost is secondary. Cost is driven usually by schedule," a director at a defense prime said in an expert call. An engineer there would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - Owning the plant pays. One listed parts maker earns a 49% gross margin in its own plants and 31% on work it farms out to partner shops (company filings). Approved incumbents earn well too: one engine-parts unit runs a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization). Aerospace programs can run about 50 years, so an approval keeps paying.
   - Selling hardware to factories has been hard. One 3D-printer maker fell from a $2.1B value to a $42.5M sale (3D Printing Journal). An AI-run machining startup in Germany went insolvent in August 2026. Buyers cap each supplier at about a quarter to a third of their work, so even a winner grows one share at a time.
7. **AI's edge today is in the work around the machine and in processes that run nonstop. Nobody has shown yet that it makes each unit cheaper as volume grows.**
   - Around the machine means quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. One startup claims it cut assembly-instruction writing from about 10 hours to 15 minutes (claim). A former employee of one parts maker said its time from order to first part fell about 50% over two years.
   - The machines themselves are underused. In a survey of more than 100 plants, cutting machines ran about 34% of the time against a 70% target. European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). Planning and control software closes that gap, with no plant to build.
   - In a plant that runs nonstop, like a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. A head of strategy at an aircraft maker put net savings from automation, after paying for the machines, at 20-30% at best. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case rests on speed and yield more than headcount.
8. **The key risk is that incumbents automate and buy faster than new companies can grow.**

- The primes decide who is approved and how much work each supplier gets. The largest aerospace parts supplier saw its share of one buyer's machining fall from about 50% to 30%, and it is now automating (expert call).
- Incumbents buy instead of build. Miners back startups through their venture arms, one paid $6.7B for a lithium producer, and a defense contractor's venture arm led a round in an AI materials startup. Drug companies are building their own AI labs, one with plans for up to $1B.
- New companies still have room. The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that base is small, old and short of people, which is where new companies can win.

2. Interesting categories

This section follows the categories in our primer. Each opens with our call, then the reasons, how the company makes money, and what has to be true for us to invest. The example companies are the ones the primer names.

A. Raw Materials

Stage A: Locating & finding minerals and mines

Investability: Economics don't work for us. We pass.

- **Why:** An explorer that does not own the mine does not own the economics. It gets a one-time sale or a small royalty, paid years later and only if someone else builds the mine. Mines average 15.7 years from discovery to production, and 19.1 years for US mines opened in 2020-24. About 40 deposits are found a year and about a fifth become mines (claim). An explorer that keeps the mine becomes a mining company: one AI explorer owns 80% of a Zambian copper project that needs about $2.3-2.5B, with first copper targeted for the early 2030s (Reuters, Ecofin). Either path is too small, too late or too capital-heavy for venture.
- **How the business works:** Explorers earn money four ways. First, an option or joint venture: a miner pays staged cash and shares and funds the drilling to earn 51-70% of the project. A typical deal pays the explorer $1M cash and $1M of shares, plus $3-5M of drilling, over 3-5 years for 60%; the explorer keeps 40% and a 2% royalty (one explorer's published terms). Second, the royalty itself, usually a net smelter return (NSR): 1-3% of metal sales after transport, often with a clause letting the miner buy part of it back for a fixed price. Third, an outright sale of early ground for a small payment plus a royalty: two 2025 sales went for C$200,000 plus a 1% royalty and US$430,000 plus a 3% royalty (company releases). Fourth, selling the royalty to a royalty company, which pays real money only once a mine is near or in production: a 1% royalty on a producing gold mine sold for US$9.9M (company release). AI explorers use the same terms. One funds up to A$4.5M of drilling on another company's ground to earn up to a 3% royalty, and plans to sell proven deposits for about 5% of the metal in the ground (claim). Software sold to miners does not repeat: one explorer ran about 80 pilots that did not turn into recurring revenue.
- **What would change our mind:** the explorer is paid in cash before a mine is built, at margins and speed a venture fund needs. No model we found does this. Miners also fund this work through their own teams and venture arms, so the field is crowded.

Example companies: Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon

Stage B: Mining & refining

Investability: Selective. Only where government debt pays for most of the plant and a buyer has signed for the output.

- **Why:** Defense and auto buyers need non-Chinese metal and magnets, and Chinese refineries commission in about 6 months against 2 to 4 years in the West. The government is closing that gap with loans and long

purchase deals. Without them, a US producer takes prices set in China and loses money: the largest US rare-earth producer lost $85.9M on $224.4M of revenue in 2025.

- **How the business works:** Miners run the mine and sell metal such as copper at market prices; one startup says its restarted Utah copper mine runs about half autonomously (claim). Refiners buy ore concentrate or scrap and sell oxides, metals or magnets under long contracts. Both own their plants and spend hundreds of millions of dollars before revenue: about 5 years to build and 3 to 5 to ramp after permits (claim).
- **What has to be true:** someone else carries most of the cost and the price risk. The deals that work look like a $620M government loan to one magnet maker, or a 10-year price floor plus a promise to buy all output of an incumbent's next magnet plant. We back the company that has both the debt and the buyer, and pass on the rest.

Example companies: Mariana Minerals, BHP, Rio Tinto, Caterpillar, Vulcan Elements, Phoenix Tailings, ReElement, Ucore, Energy Fuels, MP Materials, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

B. Manufacturing

Metal Supply

Investability: Economics don't work for us. We pass.

- **Why:** The business is crowded with established incumbents, and technology does not set a new company apart. The largest US distributor is gaining share without AI. Tariffs help US metal, but the margin stays with the incumbents.
- **How the business works:** The distributor buys metal from mills, holds it in stock, cuts it to size and sells it to shops and primes. It earns a spread on the metal plus cutting fees, and much of its cash sits in inventory. Online ordering with instant quotes already exists, so a new distributor competes on price and stock, where incumbents are bigger.
- **What would change our mind:** a new distributor shows better margins or inventory turns than the leader at scale. We have not seen this.

Example companies: Nox Metals, Reliance, Ryerson

Fabrication

Investability: Investable now.

- **Why:** Customers pay for speed, plants are cheap, and the best companies already make money. Fabrication is cutting, bending and welding metal into frames and enclosures, and assembling parts like wire harnesses, the bundles of wires that carry power and signals through an aircraft or missile. One online fabricator handles about 32,000 orders a month, is growing about 100% a year and says it has been profitable for eight years (claim). One harness maker claims it cut lead time from six months to two weeks (claim).
- **How the business works:** Customers send a design and get parts or assemblies back in days. Most companies own their plants, which cost far less than precision plants; one leased a second plant and opened it within months. Owning the plant pays: one listed parts maker earns a 49% gross margin in its own plants and 31% on work sent to partner shops (company filings).
- **What has to be true:** the company wins on speed, since one founder says he beats China on price on only about 20-25% of his work (claim), and keeps customers as they grow instead of losing them when they buy their own machines.

Example companies: SendCutSend, 1872, Senra Systems, Machina Labs, CircuitHub, Jeevy, Protolabs

Precision & Defense Hardware

Investability: Investable now, in models where someone else pays for most of the machines.

- **Why:** Primes pay a premium for fast delivery, and an approved supplier keeps the work for programs that run up to 50 years. Precision parts are metal or composite parts made to very tight limits for missiles, jets, satellites and submarines. The shops that make them are small and slow, and primes, which earn 7-12% on cost-plus contracts, outsource much of the work.
- **How the business works:** Companies sell parts under purchase orders or multi-year contracts. The factory model decides who pays for the machines. Owned factories built from scratch keep the whole margin and the data but pay before approval: one parts maker committed about $200M to a single plant, and a six-machine first plant cost "well north of $10 million" (former employee). Buying certified shops from retiring owners is cheaper, at about seven times annual earnings (claim). In a franchised network, franchisees buy the machines and the company supplies software, orders and quality rules; one claims 14 factories, 4 owned and 10 franchised (claim). Cells hosted inside a buyer's plant share the cost; one 7-year deal puts cells inside a prime's missile plant.
- **What has to be true:** the company reaches approval before the cash runs out. New plants need AS9100, the aerospace quality standard, which takes 6-12 months, then approval of each part at the buyer. We favor acquired shops, franchises and hosted cells, which need far less capital, over plants built from scratch on equity.

Example companies: Hadrian, Isembard, Divergent, Freeform, Amca, Atomic Industries, Fabri, Layup, Hybron, Rangeview, Furientis, Helicon, Westmag, Re:Build, Anduril, Precision Castparts, Howmet

C. Specialty Chemicals

Producers (hybrid)

Investability: Investable now, selectively. Back labs that already have paying drug-company work.

- **Why:** Drug companies need a US or European source for early chemistry. Most of it goes to China today. BIOSECURE, a law passed in December 2025, limits use of certain Chinese biotech suppliers, and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO). Traction is real but early. The older lab has run its automated lab since June 2025, has 152 staff, and says it works with eight of the top 20 drug companies (claim, SCI); a trade publication reported 10 big pharma clients (Citeline). The newest lab launched in October 2026 with a small number of design partners and no public customers.
- **How the business works:** The lab makes small batches of new molecules for drug teams, at milligram-to-gram scale, and does not develop its own drugs. It is paid per compound delivered, or through milestones and royalties if a molecule advances; the one deal with public terms pays this way. It owns the lab, software and reaction data. Labs cost tens of millions; one is part-funding an £89.9M expansion with about £22M of UK and Scottish grants.
- **What has to be true:** automation brings unit cost close to offshore prices, since US chemists cost several times Chinese rates, and pharma work turns into disclosed, recurring revenue. Neither lab has disclosed revenue yet.

Example companies: Chemify, Warp Sciences

D. Advanced Materials

Producers

Investability: Selective. Invest when a buyer is already qualifying the material.

- **Why:** Buyers need new or non-Chinese materials, and the company that designs and makes the material keeps the margin. China makes about 85% of battery cathode material and over 98% of the cheaper LFP type. Data centers used 415 TWh of power in 2024 and are on track for about 945 TWh by 2030, with cooling taking 7% to over 30% of it.
- **How the business works:** These companies design the material with AI in their own lab, then make it and sell it by the kilogram or ton under supply deals. Some own the plant; one claims its AI carries a material

from 1 kg a day in the lab to 1,000 kg a day in production (claim). Others use a contract maker, which cuts capex. One says its cooling fluid reached market in about 3 years, against roughly 10 years and $100M for a typical new material (claim).

- **What has to be true:** a named buyer is in qualification, which takes one to five years, and the company avoids competing on Chinese prices with an owned plant. Owned cathode plants cost hundreds of millions, and an established incumbent's battery-materials unit lost money on €436M of revenue.

Example companies: Orbital Materials, Mitra Chem, Chemix, Albemarle, Umicore, Corning

Materials Discovery Labs

Investability: Longer time horizon. More of an experiment today.

- **Why:** The labs can find materials faster, but they do not yet sell material at volume, and finding a material faster does not shorten the years customers take to qualify it. One lab made about 1,200 alloys in five or six months (claim), and an independent university test found one of its alloys beat a standard aerospace alloy. Valuations rest on the team, from a $300M seed to one reported at about $7.5B.
- **How the business works:** Labs earn today from paid R&D for partners, licensing and royalties, software licenses and government research awards of around $1M. To make material they usually go through a partner that scales it up and sells it, so the lab earns a fee or royalty unless it builds its own production.
- **What has to be true:** a lab gets a material qualified and sold at volume with a buyer, at a price that supports its valuation. One investor says materials has had "no AlphaFold moment" yet.

Example companies: Radical AI, CuspAI, Lila Sciences, Periodic Labs, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Kebotix, Citrine Informatics, Uncountable, Albert Invent, Microsoft Discovery

E. Enabling Software & Robotics

Enablers

Investability: Investable now.

- **Why:** There is no factory to build, so the money goes into product and sales, which fits a venture fund. Demand comes from the lasting problems in Part 1: plants cannot hire enough skilled people, and machines sit idle. In a survey of more than 100 plants, cutting machines ran about 34% of the time against a 70% target, and European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). The US installed 38,428 industrial robots in 2025 against 354,226 in China. In a sold-out plant each extra point of yield is close to pure profit, so the buyer can measure the payback.
- **How the business works:** Plant-control software connects to existing controls and adjusts setpoints such as a temperature or flow rate itself; one startup says it is paid on the production it adds (claim). Factory software is sold by subscription; one tool claims it cuts writing assembly instructions from about 10 hours to 15 minutes (claim). Robot companies sell or rent robots and robot cells.
- **What has to be true:** the product is priced on output or saves measurable hours, and the company wins a second and third plant after the first. The risks are slow sales to plants, incumbent automation vendors that bundle software with hardware they already installed, and robots that are not yet reliable: robot-learning success is 59-66% on short tasks against the 99.9% plus a factory needs.

Example companies: Lumaril, Arda, Mind Robotics, GrayMatter Robotics, CloudNC, AI Robotics, Bright Machines, Rapta, Allus, Faraday, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson